Website Quality & Security Audit

Target: https://www.hackthissite.org/ · Scanned on 2026-10-13T15:57:00.608616+00:00 (637.28s duration)

Overall Quality Score 50 / 100 Health benchmark based on multi-pillar automated verification	Audit Execution Status Partial Scan (89% coverage) All 4 testing pillars assessed with local tools	Total Findings 387 issues detected Consolidated into distinct vulnerability root causes

Diagnostic Pillar	Score	Coverage	Status	Findings
Functionality	**86 / 100**	50%	partial	5 issues
Security	**62 / 100**	100%	complete	88 issues
Performance	**54 / 100**	83%	partial	11 issues
UX & Accessibility	**0 / 100**	100%	complete	283 issues

Executive Summary

Mapped 8 pages; 387 deduplicated findings (252 verified by re-test). The website is experiencing issues with functionality, security, performance, and accessibility. The most critical issues are related to functionality, where the email field does not use the required type="email" attribute, and the functionality pillar has a medium severity. The security pillar has a medium severity due to missing anti-clickjacking headers and framing protection. The performance pillar has a medium severity due to slow initial HTTP responses and large contentful paint times. The accessibility pillar has a high severity due to insufficient color contrast ratios.

Key Action Priorities

• Ensure the email field uses the required type="email" attribute.

• Implement anti-clickjacking headers and framing protection to mitigate clickjacking attacks.

• Optimize the website's performance by reducing slow initial HTTP responses and large contentful paint times.

Diagnostic Telemetry & Scope Overview

Consolidated breakdown of detected issues across severity levels and testing pillars.

Severity Level	Security	Accessibility	Performance	Functionality	Total
Critical	-	-	-	-	**0**
High	-	126	2	-	**128**
Medium	40	94	9	2	**145**
Low	40	63	-	3	**106**
Info	8	-	-	-	**8**

Performance & Concurrency Telemetry

Metric	Value	Diagnostic Benchmark
Lighthouse Performance Score	**61/100**	Core web lab performance (90+ Good)
Largest Contentful Paint (LCP)	5491 ms	Perceived loading speed (<= 2500 ms good)
First Contentful Paint (FCP)	5491 ms	Initial DOM render latency (<= 1800 ms good)
Cumulative Layout Shift (CLS)	0.0076	Visual stability measure (<= 0.1 good)
Simulated Load Test (Concurrency)	**4 users / 1.0 req/s**	Avg response: 2239 ms | 95th percentile: 2500 ms

Audited Target Pages (8 discovered & verified)

URL Path	Status	Forms	Page Title / Description
https://www.hackthissite.org/	**200**	1	Hack This Site
https://www.hackthissite.org/discord	**200**	1	Hack This Site :: Discord
https://www.hackthissite.org/forums	**429**	1	Hack This Site
https://www.hackthissite.org/advertise	**200**	1	Hack This Site
https://www.hackthissite.org/user/login	**200**	1	Hack This Site :: Login
https://www.hackthissite.org/user/resetpass	**200**	2	Hack This Site :: Reset Passphrase
https://www.hackthissite.org/user/forgotusername	**200**	2	Hack This Site :: Forgot Username
https://www.hackthissite.org/donate/	**200**	1	Hack This Site

Consolidated Findings Catalog

Master inventory of distinct issues discovered during testing, grouped by root cause and ranked by severity.

#	Sev	Pillar	Issue Title	Count	Verification
1	High	UX & Accessibility	Elements must meet minimum color contrast ratio thresholds	**48x**	Verified
2	High	UX & Accessibility	Links must have discernible text	**48x**	Verified
3	High	UX & Accessibility	Images must have alternative text	**15x**	Verified
4	High	UX & Accessibility	Form elements should have a visible label	**14x**	Verified
5	High	UX & Accessibility	Frames must have an accessible name	**1x**	Verified
6	High	Performance	First visible content appears slowly	**1x**	Verified
7	High	Performance	Largest Contentful Paint is slow	**1x**	Verified
8	Medium	UX & Accessibility	All page content should be contained by landmarks	**46x**	Verified
9	Medium	UX & Accessibility	Anchor element found with no link content and no name and/or...	**15x**	Verified
10	Medium	UX & Accessibility	Img element is the only content of the link, but is missing ...	**13x**	Verified
11	Medium	UX & Accessibility	Document should have one main landmark	**8x**	Verified
12	Medium	Security	Absence of Anti-CSRF Tokens	**8x**	Observed
13	Medium	Security	CSP: Notices	**8x**	Observed
14	Medium	Security	Cross-Domain Misconfiguration	**8x**	Observed
15	Medium	Security	Sub Resource Integrity Attribute Missing	**8x**	Observed
16	Medium	Performance	Slow page response	**7x**	Verified
17	Medium	Security	Missing Anti-clickjacking Header	**5x**	Verified
18	Medium	UX & Accessibility	Page should contain a level-one heading	**4x**	Verified
19	Medium	UX & Accessibility	Heading tag found with no content. Text that is not intended...	**3x**	Verified
20	Medium	UX & Accessibility	Img element missing an alt attribute. Use the alt attribute ...	**3x**	Verified
21	Medium	Security	Framing protection is missing	**3x**	Verified
22	Medium	UX & Accessibility	Anchor element found with a valid href attribute, but no lin...	**1x**	Verified
23	Medium	UX & Accessibility	Iframe element requires a non-empty title attribute that ide...	**1x**	Verified

24	**Medium**	Functionality	Browser console or script error	**1x**	Verified
25	**Medium**	Functionality	Resource returns HTTP 429	**1x**	Verified
26	**Medium**	Performance	Slow initial HTTP response	**1x**	Verified
27	**Medium**	Performance	Slow responses under light concurrency	**1x**	Verified
28	**Low**	UX & Accessibility	Semantic markup should be used to mark emphasised or special...	**30x**	Observed
29	**Low**	UX & Accessibility	If this element contains a navigation section, it is recomme...	**15x**	Observed
30	**Low**	UX & Accessibility	Headings should not be empty	**8x**	Verified
31	**Low**	Security	Cookie No HttpOnly Flag	**8x**	Observed
32	**Low**	Security	Cookie Without Secure Flag	**8x**	Observed
33	**Low**	Security	Cookie without SameSite Attribute	**8x**	Observed
34	**Low**	Security	Cross-Domain JavaScript Source File Inclusion	**8x**	Observed
35	**Low**	UX & Accessibility	Check that this element has an inherited background colour o...	**7x**	Observed
36	**Low**	Security	X-Content-Type-Options Header Missing	**5x**	Verified
37	**Low**	Security	Content type sniffing protection is missing	**3x**	Verified
38	**Low**	UX & Accessibility	The heading structure is not logically nested. This h4 eleme...	**3x**	Observed
39	**Low**	Functionality	Expected invalid input is not rejected by native validation	**2x**	Observed
40	**Low**	Functionality	Email field does not use type="email"	**1x**	Observed
41	**Info**	Security	Modern Web Application	**8x**	Observed

Detailed Diagnostic & Remediation Cards

Comprehensive root-cause analysis, security/UX impact explanation, and code fixes for top priority issues.

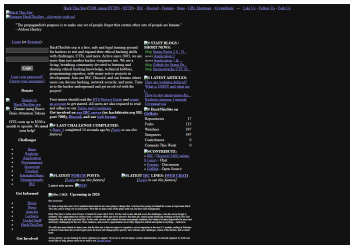
Issue #1: Elements must meet minimum color contrast ratio thresholds

Pillar: UX & Accessibility | Severity: High | Occurrences: 48 detected | Sources: axe-core

Technical Explanation & Impact:
Ensure the contrast between foreground and background colors meets WCAG 2 AA minimum contrast ratio thresholds

Remediation Guidance:
Correct the affected element according to the color-contrast rule, then re-run axe-core. Guidance: https://dequeuniversity.com/rules/axe/4.10/color-contrast?application=axeAPI

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)
Sample Selectors: <code>#active</code>, <code>#topbar > a:nth-child(2)</code>, <code>#topbar > a:nth-child(3)</code>



Evidence: Page screenshot; selector identifies the affected element.

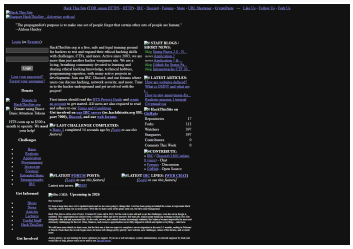
Issue #2: Links must have discernible text

Pillar: UX & Accessibility | Severity: High | Occurrences: 48 detected | Sources: axe-core

Technical Explanation & Impact:
Ensure links have discernible text

Remediation Guidance:
Correct the affected element according to the link-name rule, then re-run axe-core. Guidance: https://dequeuniversity.com/rules/axe/4.10/link-name?application=axeAPI

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)
Sample Selectors: <code>ul:nth-child(8) > li:nth-child(7) > .nav[href=""]</code>, <code>ul:nth-child(10) > li:nth-child(7) > .nav[href=""]</code>, <code>li:nth-child(12) > .nav[href=""]</code>



Evidence: Page screenshot; selector identifies the affected element.

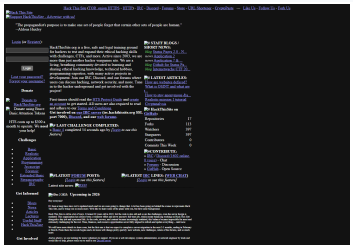
Issue #3: Images must have alternative text

Pillar: UX & Accessibility | Severity: High | Occurrences: 15 detected | Sources: axe-core

Technical Explanation & Impact:
Ensure <img> elements have alternative text or a role of none or presentation

Remediation Guidance:
Correct the affected element according to the image-alt rule, then re-run axe-core. Guidance: https://dequeuniversity.com/rules/axe/4.10/image-alt?application=axeAPI

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)
Sample Selectors: <code>img[border="0"]:nth-child(2)</code>, <code>img[border="0"]:nth-child(2)</code>, <code>img[src$="advert_options.jpg"]</code>



Evidence: Page screenshot; selector identifies the affected element.

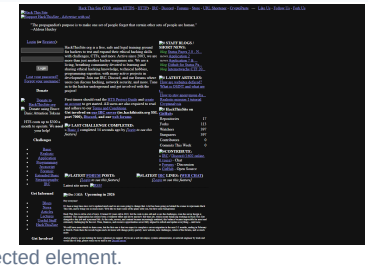
Issue #4: Form elements should have a visible label

Pillar: UX & Accessibility | Severity: High | Occurrences: 14 detected | Sources: axe-core

Technical Explanation & Impact:
Ensure that every form element has a visible label and is not solely labeled using hidden labels, or the title or aria-describedby attributes

Remediation Guidance:
Correct the affected element according to the label-title-only rule, then re-run axe-core. Guidance: https://dequeuniversity.com/rules/axe/4.10/label-title-only?application=axeAPI

Affected Locations: Detected on 7 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)
Sample Selectors: <code>input[type="text"]</code>, <code>input[type="password"]</code>, <code>input[type="text"]</code>



Evidence: Page screenshot; selector identifies the affected element.

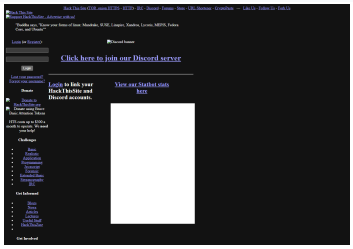
Issue #5: Frames must have an accessible name

Pillar: UX & Accessibility | Severity: High | Occurrences: 1 detected | Sources: axe-core

Technical Explanation & Impact:
Ensure <iframe> and <frame> elements have an accessible name

Remediation Guidance:
Correct the affected element according to the frame-title rule, then re-run axe-core. Guidance: https://dequeuniversity.com/rules/axe/4.10/frame-title?application=axeAPI

Affected Locations: Detected on 1 page(s) (https://www.hackthissite.org/discord)
Sample Selectors: <code>iframe</code>



Evidence: Page screenshot; selector identifies the affected element.

Issue #6: First visible content appears slowly

Pillar: Performance | Severity: High | Occurrences: 1 detected | Sources: Lighthouse

Technical Explanation & Impact:
This Lighthouse laboratory measurement is outside the recommended range. It is not a real-user percentile and may be affected by blocked third-party assets and container resources.

Remediation Guidance:
Reduce blocking CSS, improve server response time, and deliver critical above-the-fold assets earlier.

Affected Locations: Detected on 1 page(s) (https://www.hackthissite.org/)



Evidence: Lighthouse mobile final screenshot.

Issue #7: Largest Contentful Paint is slow

Pillar: Performance | Severity: High | Occurrences: 1 detected | Sources: Lighthouse

Technical Explanation & Impact:
This Lighthouse laboratory measurement is outside the recommended range. It is not a real-user percentile and may be affected by blocked third-party assets and container resources.

Remediation Guidance:
Optimize the main content image, preload critical resources, cache the first response, and reduce render-blocking resources.

Affected Locations: Detected on 1 page(s) (https://www.hackthissite.org/)



Evidence: Lighthouse mobile final screenshot.

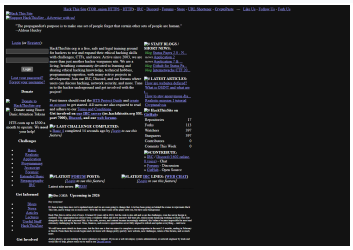
Issue #8: All page content should be contained by landmarks

Pillar: UX & Accessibility | Severity: Medium | Occurrences: 46 detected | Sources: axe-core

Technical Explanation & Impact:
Ensure all page content is contained by landmarks

Remediation Guidance:
Correct the affected element according to the region rule, then re-run axe-core. Guidance: https://dequeuniversity.com/rules/axe/4.10/region?application=axeAPI

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)
Sample Selectors: <code>#topbar</code>, <code>.hts-header</code>, <code>.siteheader > tbody > tr:nth-child(1)</code>



Evidence: Page screenshot; selector identifies the affected element.

Issue #9: Anchor element found with no link content and no name and/or ID attribute.

Pillar: UX & Accessibility | Severity: Medium | Occurrences: 15 detected | Sources: Pa11y / HTML_CodeSniffer

Technical Explanation & Impact:
Anchor element found with no link content and no name and/or ID attribute.

Remediation Guidance:
Correct the semantic markup or style described by this WCAG rule and verify with Pa11y plus keyboard testing.

Affected Locations: Detected on 3 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/discord, https://www.hackthissite.org/forums)
Sample Selectors: <code>html > body > table > tbody > tr:nth-child(2) > td > table > tbody > tr > td:nth-child(1) > div > ul:nth-child(8) > li:nth-child(7) > a</code>, <code>html > body > table > tbody > tr:nth-child(2) > td > table > tbody > tr > td:nth-child(1) > div > ul:nth-child(10) > li:nth-child(7) > a</code>, <code>html > body > table > tbody > tr:nth-child(2) > td > table > tbody > tr > td:nth-child(1) > div > ul:nth-child(12) > li:nth-child(12) > a</code>

Issue #10: Img element is the only content of the link, but is missing alt text. The alt text should describe the purpose of the link.

Pillar: UX & Accessibility | Severity: Medium | Occurrences: 13 detected | Sources: Pa11y / HTML_CodeSniffer

Technical Explanation & Impact:
Img element is the only content of the link, but is missing alt text. The alt text should describe the purpose of the link.

Remediation Guidance:
Correct the semantic markup or style described by this WCAG rule and verify with Pa11y plus keyboard testing.

Affected Locations: Detected on 3 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/discord, https://www.hackthissite.org/forums)
Sample Selectors: <code>html > body > table > tbody > tr:nth-child(2) > td > table > tbody > tr > td:nth-child(1) > div > a:nth-child(18)</code>, <code>html > body > table > tbody > tr:nth-child(2) > td > table > tbody > tr > td:nth-child(1) > div > a:nth-child(18)</code>, <code>html > body > div:nth-child(7) > p > a:nth-child(1)</code>

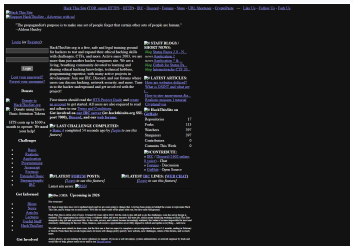
Issue #11: Document should have one main landmark

Pillar: UX & Accessibility | Severity: Medium | Occurrences: 8 detected | Sources: axe-core

Technical Explanation & Impact:
Ensure the document has a main landmark

Remediation Guidance:
Correct the affected element according to the landmark-one-main rule, then re-run axe-core. Guidance: https://dequeuniversity.com/rules/axe/4.10/landmark-one-main?application=axeAPI

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)
Sample Selectors: <code>html</code>, <code>html</code>, <code>html</code>



Evidence: Page screenshot; selector identifies the affected element.

Issue #12: Absence of Anti-CSRF Tokens

Pillar: Security | Severity: Medium | Occurrences: 8 detected | Sources: OWASP ZAP passive baseline

Technical Explanation & Impact:
No Anti-CSRF tokens were found in a HTML submission form. A cross-site request forgery is an attack that involves forcing a victim to send an HTTP request to a target destination without their knowledge or intent in order to perform an action as the victim. This remains an unconfirmed signal, not a verified exploit.

Remediation Guidance:
Add Anti-CSRF tokens to the login form.

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)

Issue #13: CSP: Notices

Pillar: Security | Severity: Medium | Occurrences: 8 detected | Sources: OWASP ZAP passive baseline

Technical Explanation & Impact:
Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain types of attacks. Including Cross Site Scripting (XSS) and data injection attacks. These attacks are used for everything from phishing to stealing sensitive information. This remains an unconfirmed signal, not a verified exploit.

Remediation Guidance:
Implement CSP headers to restrict the sources of scripts and styles that can be loaded on the page.

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)
Sample Selectors: <code>Content-Security-Policy</code>, <code>Content-Security-Policy</code>, <code>Content-Security-Policy</code>

Issue #14: Cross-Domain Misconfiguration

Pillar: Security | Severity: Medium | Occurrences: 8 detected | Sources: OWASP ZAP passive baseline

Technical Explanation & Impact:
Web browser data loading may be possible, due to a Cross Origin Resource Sharing (CORS) misconfiguration on the web server. This can be used for cross-site request forgery and other types of attacks that rely on cross-origin requests. This remains an unconfirmed signal, not a verified exploit.

Remediation Guidance:
Configure CORS properly on the web server to restrict the origins that can access the page.

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)

Issue #15: Sub Resource Integrity Attribute Missing

Pillar: Security | Severity: Medium | Occurrences: 8 detected | Sources: OWASP ZAP passive baseline

Technical Explanation & Impact:
The integrity attribute is missing on a script or link tag served by an external server. The integrity tag prevents an attacker who has gained access to this server from injecting a malicious content. This remains an unconfirmed signal, not a verified exploit.

Remediation Guidance:
Add the integrity attribute to the script or link tag.

Affected Locations: Detected on 8 page(s) (https://www.hackthissite.org/, https://www.hackthissite.org/advertise, https://www.hackthissite.org/discord...)

Additional Low & Informational Findings

Secondary observations and hygiene recommendations summarized for quick review.

Issue Title	Pillar	Sev	Count	Remediation Summary
Slow page response	Performance	Medium	**7x**	Profile server-side work for this route, cache expensive results, and avoid bloc...
Missing Anti-clickjacking Header	Security	Medium	**5x**	Modern Web browsers support the Content-Security-Policy and X-Frame-Options HTTP...
Page should contain a level-one heading	UX & Accessibility	Medium	**4x**	Correct the affected element according to the page-has-heading-one rule, then re...
Heading tag found with no content. Text that ...	UX & Accessibility	Medium	**3x**	Correct the semantic markup or style described by this WCAG rule and verify with...
Img element missing an alt attribute. Use the...	UX & Accessibility	Medium	**3x**	Correct the semantic markup or style described by this WCAG rule and verify with...
Framing protection is missing	Security	Medium	**3x**	Implement clickjacking protection measures....
Anchor element found with a valid href attrib...	UX & Accessibility	Medium	**1x**	Correct the semantic markup or style described by this WCAG rule and verify with...
Iframe element requires a non-empty title att...	UX & Accessibility	Medium	**1x**	Correct the semantic markup or style described by this WCAG rule and verify with...

Browser console or script error	Functionality	Medium	**1x**	Reproduce in the browser console, fix the failing script or request, and verify ...
Resource returns HTTP 429	Functionality	Medium	**1x**	Repair the endpoint or asset path and ensure the request returns the intended re...
Slow initial HTTP response	Performance	Medium	**1x**	Measure server-side latency, cache appropriate responses, optimize slow queries,...
Slow responses under light concurrency	Performance	Medium	**1x**	Profile request queues, database latency and cache misses. Repeat from a represe...
Semantic markup should be used to mark emphas...	UX & Accessibility	Low	**30x**	Correct the semantic markup or style described by this WCAG rule and verify with...
If this element contains a navigation section...	UX & Accessibility	Low	**15x**	Correct the semantic markup or style described by this WCAG rule and verify with...
Headings should not be empty	UX & Accessibility	Low	**8x**	Correct the affected element according to the empty-heading rule, then re-run ax...
Cookie No HttpOnly Flag	Security	Low	**8x**	Enable the HttpOnly flag for all cookies....
Cookie Without Secure Flag	Security	Low	**8x**	Enable the secure flag for all cookies....
Cookie without SameSite Attribute	Security	Low	**8x**	Enable the SameSite attribute for all cookies....
Cross-Domain JavaScript Source File Inclusion	Security	Low	**8x**	Ensure that all script files are from trusted domains....
Check that this element has an inherited back...	UX & Accessibility	Low	**7x**	Correct the semantic markup or style described by this WCAG rule and verify with...
X-Content-Type-Options Header Missing	Security	Low	**5x**	Ensure that the application/web server sets the Content-Type header appropriatel...
Content type sniffing protection is missing	Security	Low	**3x**	Add `x-content-type-options: nosniff` to the headers....
The heading structure is not logically nested...	UX & Accessibility	Low	**3x**	Correct the semantic markup or style described by this WCAG rule and verify with...
Expected invalid input is not rejected by nat...	Functionality	Low	**2x**	Confirm the field's intended rules, provide accessible inline feedback, and enfo...
Email field does not use type="email"	Functionality	Low	**1x**	Use <input type="email" required> and keep server-side validation as the authori...
Modern Web Application	Security	Info	**8x**	Use the Client Spider for automated exploration....

Diagnostic Engine Audit Trail & Methodology

Execution verification log for all automated scanners and local intelligence models.

Pipeline Step	Engine / Tool	Status	Execution Detail
Authorization & network safety	Internal	**completed**	Exact origin approved for the scan.
Crawl & map the website	Internal	**partial**	Mapped 8 pages. 0 page failures. Same-origin links only; no mutation s
AI page classification & flows	Internal	**completed**	Local classification/flow inference completed 5/5 bounded tasks. 3 kno
AI adaptive diagnostic plan	Internal	**completed**	Local AI proposed 8 safe, page-aware recommendations; supported checks
AI cases & safe form validation	Internal	**partial**	Executed 16/16 client-side cases; 0 failed and 0 were skipped. No cred
Internal links & browser errors	Internal	**partial**	Checked 18/20 internal links; 2 request failures and 0 policy blocks.
Headers, cookies & inert probes	Internal	**completed**	Read-only headers, cookies, file signatures; probed 0 observed GET par
OWASP ZAP passive baseline	Internal	**completed**	Passive baseline over safe crawl seeds; active scan and autonomous spi
Nuclei read-only templates	Internal	**completed**	Only 3 committed GET-only templates; no fuzzing, OAST or template down
HTTP response measurements	Internal	**partial**	Three sequential GET measurements through the safety proxy. Plus per-p
Lighthouse performance audit	Internal	**completed**	Mobile laboratory Lighthouse. LCP/CLS/TBT/FCP measured; field INP not
Capped Locust concurrency test	Internal	**completed**	GET-only load: 4 users / 10s, capped at 20 / 30 with early-stop protec
axe-core accessibility audit	Internal	**completed**	axe-core audited 8 pages; 0 pages could not be audited.
Pa11y independent WCAG checks	Internal	**completed**	Pa11y audited 3 pages; 0 failures.
AI desktop & mobile visual review	Internal	**completed**	Local vision review completed 4/4 screenshot tasks; visual findings st
Re-test & deterministic deduplication	Internal	**completed**	Deterministic re-tests applied; matching root causes merged with prove
AI security finding triage	Internal	**completed**	Local triage fully completed 4/4 batches; 0 batches applied partial kn
AI final report & suggested fixes	Internal	**completed**	Local AI synthesis applied; evidence, verification and scores remain d

Scope, Safe Harbor & Limitations

• URL-only black-box snapshot, not a comprehensive penetration test or WCAG certification.

• No account creation, credential attempts, payment, upload, mutation method or destructive action is submitted. Server-side login/checkout coverage is intentionally limited.

• Only same-origin resources are loaded. Third-party scripts/fonts/images are blocked; rendering and performance may differ from a normal browser.

• Laboratory LCP/CLS and TBT are measured. Field Core Web Vitals percentiles and INP are not invented.

• Some local AI tasks were unavailable or failed; no external/closed-model fallback was used.

• Crawl & map the website: partial. Mapped 8 pages. 0 page failures. Same-origin links only; no mutation submissions.

• AI cases & safe form validation: partial. Executed 16/16 client-side cases; 0 failed and 0 were skipped. No credentials or state-changing submissions; server-side login/checkout behavior remains untested. Some cases used deterministic fallback, not AI generation.

• Internal links & browser errors: partial. Checked 18/20 internal links; 2 request failures and 0 policy blocks. SPA fragments require browser coverage. Replayed 3/3 AI-selected known-page flows through read-only navigation; 3 completed all observations.

• HTTP response measurements: partial. Three sequential GET measurements through the safety proxy. Plus per-page timing and sub-resource sizes for 8 pages (1 resources).

Methodology & Safe Harbor Disclaimer: This assessment was executed using safe, read-only diagnostic tools in accordance with automated quality testing guidelines. Reported findings reflect observed software behavior at the time of testing and do not constitute an exhaustive warranty or formal penetration certification.